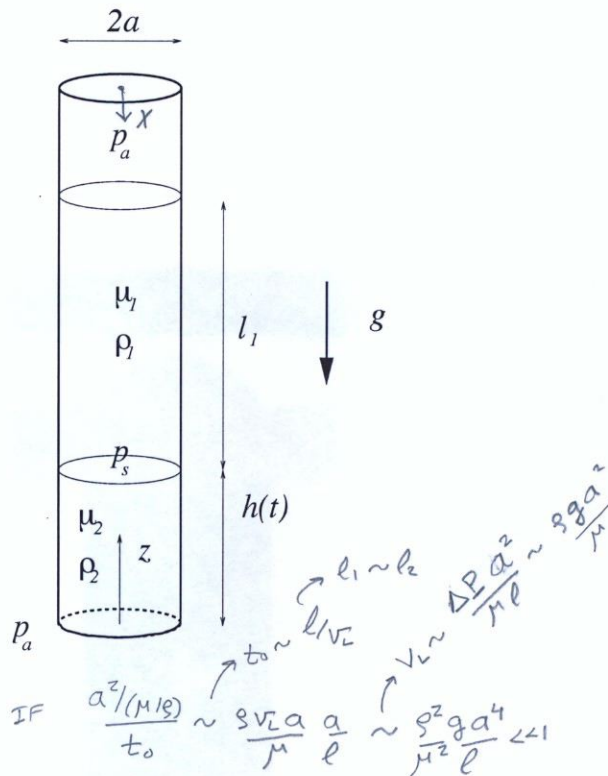


VFS

A vertical tube of radius a open at the top contains initially two columns of heights l_1 and l_2 of two immiscible liquids with different densities ρ_1 and ρ_2 and viscosities μ_1 and μ_2 . At a given instant of time, the bottom end of the pipe opens up, releasing the fluid 2 to the atmosphere. During the discharge process, determine the volume flux as a function of time as well as the pressure distribution along the pipe. Calculate the time for complete discharge of the bottom liquid column.



THE FLOW IS DOMINATED BY VISCOSITY IF $\frac{a^2/(M/g)}{t_0} \sim \frac{8 \sqrt{g} a}{\mu} \frac{a}{l} \sim \frac{8 g^2 a^4}{\mu^2 l} \ll 1$

TOP FLUID $V_x = \frac{P_l}{4\mu_1} (a^2 - r^2), Q_1 = \frac{\pi}{8\mu_1} a^4 P_l = \frac{\pi}{8\mu_1} a^4 (P_a - P_s + \rho_1 g l_1)$

BOTTOM FLUID $Q_2 = \frac{\pi}{8\mu_2} a^4 P_2 = \frac{\pi}{8\mu_2} a^4 \frac{P_s - P_a + \rho_2 g h}{h}$

$Q_1 = Q_2 = Q \Rightarrow P_a - P_s + \rho_1 g l_1 = \frac{8}{\pi} \frac{\mu_1 l_1}{a^4} Q$

$P_s - P_a + \rho_2 g h = \frac{8}{\pi} \frac{\mu_2 h}{a^4} Q$

$(\rho_1 l_1 + \rho_2 h) g = \frac{8}{\pi a^4} (\mu_1 l_1 + \mu_2 h) Q = - \frac{8}{a^2} (\mu_1 l_1 + \mu_2 h) \frac{dh}{dt}$

$$\int_{l_2}^h \frac{(\mu_1 l_1 + \mu_2 h) dh}{(\rho_1 l_1 + \rho_2 h)} = - \frac{g a^2}{8} \int_0^t dt$$



$dV = \pi a^2 dh$